TOPSTECHNOLOGIES
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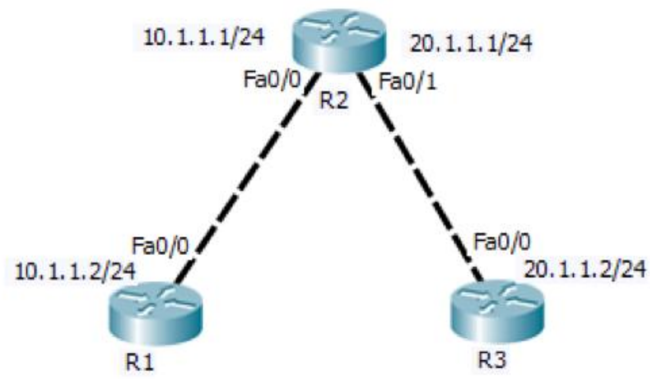
Day 4

Introduction to OSPF Theory : How OSPF works: Areas, LSAs, and cost . Practical: Observe OSPF adjacency formation in Packet Tracer. Task: Write down steps to configure OSPF in a network.

1. Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links.

ANS :

- Network Topology



2. Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.

ANS:

- **Assign IP :**

Router	Interface	IP Address	Subnet Mask
R1	FastEthernet0/0	**10.1.1.2**	**255.255.255.0**
R2	FastEthernet0/0	**10.1.1.1**	**255.255.255.0**
R2	FastEthernet0/1	**20.1.1.1**	**255.255.255.0**
R3	FastEthernet0/0	**20.1.1.2**	**255.255.255.0**

- **PING Command :**

  From Router R1
  ping 10.1.1.1
  Result: Success (100% success rate)

3. Configure OSPF on each router using the router ospf 1 command and the network command to advertise all directly connected networks.

ANS :

- **Router R1 Configuration**

```
enable
configure terminal

router ospf 1
network 10.1.1.0 0.0.0.255 area 0
```

- **Router R2 Configuration**

```
enable
configure terminal

router ospf 1
network 10.1.1.0 0.0.0.255 area 0
network 20.1.1.0 0.0.0.255 area 0
```

- **Router R3 Configuration**

```
enable
configure terminal

router ospf 1
network 20.1.1.0 0.0.0.255 area 0
```

- **Verify OSPF Configuration**

  Run the following command on all three routers:
  show ip protocols
  Expected Output:
  - Routing Protocol: **OSPF**
  - Process ID: **1**
  - Area: **0**
  - Advertised Networks:
    - **R1:** 10.1.1.0/24
    - **R2:** 10.1.1.0/24, 20.1.1.0/24
    - **R3:** 20.1.1.0/24

4. Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router. Hint : Look for the FULL/DR or FULL/BDR state to confirm adjacency.

ANS:

- **Command**

  Run the following command on each router:
  show ip ospf neighbor

- **Router R1**

  R1# show ip ospf neighbor

  Neighbor ID Pri State Dead Time Address Interface
  2.2.2.2 1 FULL/DR 00:00:39 10.1.1.1 FastEthernet0/0

- **Router R2**

  R2# show ip ospf neighbor

  Neighbor ID Pri State Dead Time Address Interface
  1.1.1.1 1 FULL/BDR 00:00:36 10.1.1.2 FastEthernet0/0
  3.3.3.3 1 FULL/DR 00:00:34 20.1.1.2 FastEthernet0/1

- **Router R3**

  R3# show ip ospf neighbor

  Neighbor ID Pri State Dead Time Address Interface
  2.2.2.2 1 FULL/BDR 00:00:38 20.1.1.1 FastEthernet0/0

- **Observation**

Router	Neighbor	State	Interface
R1	R2	**FULL/DR**	FastEthernet0/0
R2	R1	**FULL/BDR**	FastEthernet0/0
R2	R3	**FULL/DR**	FastEthernet0/1
R3	R2	**FULL/BDR**	FastEthernet0/0

5. Change the OSPF cost on one interface using the ip ospf cost command and verify how it affects the OSPF routing table. Hint : Use show ip route ospf to see route changes after modifying the cost.

ANS :

- **Step 1: Change the OSPF Cost**

  On **Router R2**, increase the OSPF cost of the interface connected to **R3**.
  R2> enable
  R2# configure terminal
  R2(config)# interface FastEthernet0/1
  R2(config-if)# ip ospf cost 100
  R2(config-if)# end

- **Step 2: Verify the OSPF Routing Table**

  Run the following command on **Router R1**:
  show ip route ospf

- **Sample Output**

  R1# show ip route ospf

  O 20.1.1.0/24 [110/101] via 10.1.1.1, FastEthernet0/0
  Run the same command on **Router R3**:
  R3# show ip route ospf

  O 10.1.1.0/24 [110/101] via 20.1.1.1, FastEthernet0/0

- **Observation Table**

Before Changing Cost	**After Changing Cost**
OSPF cost was the default value (1).	OSPF cost increased to **100** on FastEthernet0/1.
Route metric was lower.	Route metric became higher (**101**).
Traffic used the only available path through R2.	Traffic still used the same path because no alternate route exists.
OSPF routing table showed a lower metric.	OSPF routing table showed a higher metric.